Question			Answer	Marks	AO	Guidance
14	(a)		The size and shape of the ball are neglected	B1 [1]	2.4	Sensible comment equivalent to "the object is modelled as a point mass". Do not allow for a statement that includes correct and incorrect ideas. (see appendix)
14	(b)		Vertical motion $s = -3.6$, $a = -9.8$, $t = 2.4$ $s = u_y t + \frac{1}{2}at^2$ $-3.6 = 2.4u_y - 4.9 \times 2.4^2$ Giving $u_y = 10.26$ m s^{-1}	 M1 A1 [2]	 3.1b 1.1	Use of *suvat* equation(s) with $s = \pm 3.6$ and $v \neq 0$ leading to a value for u_y Allow sign errors cao Allow SC1 if M0 awarded and $u \sin \alpha$ seen. Or award SC2 for $u \sin \alpha$ in (b) if 10.26 seen in (c)
14	(c)		The horizontal velocity of the ball must be the same as the velocity of the man $u = \sqrt{4^2 + 10.26^2} = 11.01$ $\alpha = \tan^{-1}\frac{10.26}{4} = 68.7°$ **Alternative solution for last 3 marks** $-3.6 = 2.4\frac{4}{\cos\alpha}\sin\alpha - 4.9 \times 2.4^2$ $\alpha = 68.7°$ So $u = \left[\frac{4}{\cos 68.7}\right] = 11.01$	B1 B1 M1 A1 M1 A1 B1 [4]	3.1b 1.1 1.1 1.1 	Soi eg $u \cos \alpha = 4$ FT their (b) and their horizontal velocity Also allow from solving $u \cos \alpha = 4$ or $u \sin \alpha = 10.26$ FT their (b) and their horizontal velocity Substitute for u in terms of $\cos \alpha$ leading to a value for $\tan \alpha$ FT their horizontal velocity FT their value for α

Exemplar responses for Q14(a)

Response	Mark
The mass is concentrated at the centre	**B0**
The object has no mass	**B0**
There is no air resistance	**B0**
The weight acts at the centre [of mass]	**B0**
There is no spin	**B1**
The ball's size and shape do not matter	**B1**
Only its mass is taken into account so it doesn't spin	**B1**
The size of the ball is negligible and there are no external forces acting on the ball	**B0**
Point mass means there is no air resistance	**B0**
There is no air resistance because the object has no size	**B0**